

Day 1(LOS)

Part A — User and System Investigation (10 minutes)

1. Determine the username of the account you are currently using.

```
srinivas@DESKTOP-E4HFS1K:~$ whoami
srinivas
srinivas@DESKTOP-E4HFS1K:~$
```

2. Display your UID, primary GID, and supplementary groups.

```
alice@DESKTOP-E4HFS1K:~$ id
uid=1003(alice) gid=1003(alice) groups=1003(alice),100(users)
alice@DESKTOP-E4HFS1K:~$
```

3. Display all users currently logged in to the system.

```
srinivas@DESKTOP-E4HFS1K:~$ who
srinivas  seat0  2026-09-01 09:43 (:0)
srinivas@DESKTOP-E4HFS1K:~$ █
```

4. Determine what activities the currently logged-in users are performing.

```
srinivas@DESKTOP-E4HFS1K:~$ w
19:56:06 up 6 min, 0 user, load average: 0.52, 0.58, 0.59
USER      TTY      FROM          LOGIN@      IDLE       JCPU       PCPU       WHAT
srinivas@DESKTOP-E4HFS1K:~$ █
```

5. Examine /etc/passwd and identify the following information for your current account: Username

UID

GID

Home directory

Login shell

```
root@DESKTOP-E4HFS1K:~# grep -E '^(security|developers):' /etc/group
security:x:1005:alice,bob
developers:x:1006:msis
```

6. Determine how many user-account entries are present in /etc/passwd.

```
srinivas@DESKTOP-E4HFS1K:~$ wc -l /etc/passwd
30 /etc/passwd
srinivas@DESKTOP-E4HFS1K:~$
```

7. Display only the username and UID fields from /etc/passwd.

```
srinivas@DESKTOP-E4HFS1K:~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:998
systemd-timesync:996
dhcpd:100
messagebus:101
syslog:102
systemd-resolve:991
uidd:103
landscape:104
polkitd:990
srinivas:1000
testuser:1001
msis:1002
srinivas@DESKTOP-E4HFS1K:~$
```

8. Find all accounts whose UID is 0. Explain the security significance of UID 0. Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

```
srinivas@DESKTOP-E4HFS1K:~$ awk -F: '$3 == 0 {print $1, $3}' /etc/passwd
root 0
srinivas@DESKTOP-E4HFS1K:~$
```

Explanation:

UID 0 represents the superuser (root). Any account with UID 0 has full administrative privileges. Therefore, unexpected accounts with UID 0 are a major security risk because they can control the entire system.

```
srinivas@DESKTOP-E4HFS1K:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
```

```
root@DESKTOP-E4HFS1K:~# sudo cat /etc/shadow
root:$y$j9T$1KwUCXtIRtjItP0oTNwN0$gV584DS1nAODx.Ff0xMhGoaGe3k05E950Nmp5YFZXD:20687:0:99999:7:::
daemon:*:20094:0:99999:7:::
bin:*:20094:0:99999:7:::
sys:*:20094:0:99999:7:::
sync:*:20094:0:99999:7:::
games:*:20094:0:99999:7:::
man:*:20094:0:99999:7:::
lp:*:20094:0:99999:7:::
mail:*:20094:0:99999:7:::
news:*:20094:0:99999:7:::
uucp:*:20094:0:99999:7:::
proxy:*:20094:0:99999:7:::
www-data:*:20094:0:99999:7:::
backup:*:20094:0:99999:7:::
list:*:20094:0:99999:7:::
irc:*:20094:0:99999:7:::
_apt:*:20094:0:99999:7:::
nobody:*:20094:0:99999:7:::
systemd-networkd:!*:20094:!!!!:
systemd-timesyncd:!*:20094:!!!!:
dhcpcd:!*:20094:!!!!:
messagebus:!*:20094:!!!!:
syslog:!*:20094:!!!!:
systemd-resolve:!*:20094:!!!!:
uuidd:!*:20094:!!!!:
landscape:!*:20094:!!!!:
polkitd:!*:20094:!!!!:
srinivas:$y$j9T$2G2rxGb6DK8DAp7gItWsh1$gOV4qJkSYqvwyuQgZgL28f.sgYRZW6Guw/Zs8LfQcS1:20687:0:99999:7:::
testuser:$y$j9T$eoH6/Mg1qz0uIZC1098E3.$z33sRt8ELntG7woDLlWtAXG19StyPKBH9.40szWsc:20697:0:99999:7:::
msis:$y$j9T$MASEjdoVtzi5Jd/FXmstj/$T5IjgdTULDAay8PrEipZM9/AeqHxJmq/Qnh9ZBmBI7:20697:0:99999:7:::
root@DESKTOP-E4HFS1K:~#
```

Answer:

/etc/passwd:

Contains user account information.

Contains username, UID, GID, home directory, and login shell.

Usually readable by all users.

/etc/shadow:

Contains password hashes.

Contains password expiration and aging information.

Accessible only to root or authorized users.

Why are password hashes stored in /etc/shadow?

Password hashes are stored in /etc/shadow for security. If hashes were publicly readable, attackers could copy them and attempt to crack passwords offline.

Part B — Creating and Verifying Users (20 Minutes)

1. Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

The -m option creates a home directory.

```
root@DESKTOP-E4HFS1K:~# adduser alice
info: Adding user `alice' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `alice' (1003) ...
info: Adding new user `alice' (1003) with group `alice (1003)' ...
info: Creating home directory `/home/alice' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N] y
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
info: Adding new user `alice' to supplemental / extra groups `users' ...
info: Adding user `alice' to group `users' ...
root@DESKTOP-E4HFS1K:~# adduser bob
info: Adding user `bob' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `bob' (1004) ...
info: Adding new user `bob' (1004) with group `bob (1004)' ...
info: Creating home directory `/home/bob' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bob
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
info: Adding new user `bob' to supplemental / extra groups `users' ...
info: Adding user `bob' to group `users' ...
root@DESKTOP-E4HFS1K:~# grep -E '^(alice|bob|charlie):' /etc/passwd
```

2. Verify that all three accounts have been added to /etc/passwd.

```
root@DESKTOP-E4HFS1K:~# grep -E '^(security|developers):' /etc/group
security:x:1005:alice,bob
developers:x:1006:msis
```

3. Determine the UID assigned to each newly created user.

```
root@DESKTOP-E4HFS1K:~# id -u alice
1003
root@DESKTOP-E4HFS1K:~# id -u bob
1004
```

4. Determine the home directory and default shell assigned to alice.

Find where Alice's personal files are stored and which shell she uses when logging in.

```
root@DESKTOP-E4HFS1K:~# getent passwd alice
alice:x:1003:1003:,,,:/home/alice:/bin/bash
root@DESKTOP-E4HFS1K:~#
```

Answer:

Home directory: /home/alice

Default shell: /bin/bash

5. Inspect /home and verify that the required home directories were created.

To check if directories were created for all three users.

```
root@DESKTOP-E4HFS1K:~# ls -l /home/
total 0
drwxr-x--- 1 alice    alice    4096 Sep  1 20:27 alice
drwxr-x--- 1 bob      bob      4096 Sep  1 20:30 bob
drwxr-x--- 1 msis     msis     4096 Sep  1 19:35 msis
drwxr-x--- 1 srinivas srinivas 4096 Aug 25 18:14 srinivas
drwxr-x--- 1 testuser testuser 4096 Sep  1 19:13 testuser
root@DESKTOP-E4HFS1K:~#
```

6. Set passwords for all three accounts.

Assign login passwords to the newly created users.

```
root@DESKTOP-E4HFS1K:~# passwd alice
New password:
Retype new password:
passwd: password updated successfully
root@DESKTOP-E4HFS1K:~#
```

7. Switch from your current account to alice using a login shell.

Log in as Alice while loading her normal login environment.

```
root@DESKTOP-E4HFS1K:~# su - alice
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 4.4.0-19041-Microsoft x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep  1 20:43:34 IST 2026

System load:            0.52
Usage of /home:         unknown
Memory usage:          77%
Swap usage:            4%
Processes:             10
Users logged in:       0
IPv4 address for wifi0: 10.114.118.252
IPv6 address for wifi0: 2409:40f2:301f:ee3d:205b:1d98:8e21:eeaf
IPv6 address for wifi0: 2409:40f2:301f:ee3d:cde8:4a9a:921c:8ccd

=> /mnt/e is using 94.3% of 48.12GB

This message is shown once a day. To disable it please create the
/home/alice/.hushlogin file.
alice@DESKTOP-E4HFS1K:~$ whoami
alice
```

Explanation:

The - creates a login shell and loads Alice's login environment, including her home directory and environment settings.

8. Verify the change using an appropriate command, and then return to your original account.

```
srinivas@DESKTOP-E4HFS1K:~$ id
uid=1000(srinivas) gid=1000(srinivas) groups=1000(srinivas),100(users)
srinivas@DESKTOP-E4HFS1K:~$
```

Part C — Group Management and Access Control (30 Minutes)

1. Create two groups: security and developers

Create two groups that can be used to organise users and control access

```
root@DESKTOP-E4HFS1K:~# groupadd security
root@DESKTOP-E4HFS1K:~# groupadd developers
```

2. Add alice and charlie to the security group.

Make two users supplementary members of the security group

Usermod modifies an existing user

-G specifies supplementary groups

-a appends the group and preserves existing group memberships

3. Add msis to the developers group.

```
root@DESKTOP-E4HFS1K:~# usermod -aG security alice
root@DESKTOP-E4HFS1K:~# usermod -aG security bob
root@DESKTOP-E4HFS1K:~# usermod -aG developers msis
```

4. Without switching accounts, verify the group membership of all three users.

```
root@DESKTOP-E4HFS1K:~# id alice
uid=1003(alice) gid=1003(alice) groups=1003(alice),100(users),1005(security)
root@DESKTOP-E4HFS1K:~# id msis
uid=1002(msis) gid=1002(msis) groups=1002(msis),100(users),1006(developers)
```

5. Display the entries corresponding to security and developers from /etc/group.

```
root@DESKTOP-E4HFS1K:~# grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1003:1003:,,,:/home/alice:/bin/bash
bob:x:1004:1004:,,,:/home/bob:/bin/bash
```

6. Add bob to the security group without removing him from the developers group.

```
root@DESKTOP-E4HFS1K:~# usermod -aG security msis
root@DESKTOP-E4HFS1K:~# groups msis
```

7. Verify that bob now belongs to both groups.

```
root@DESKTOP-E4HFS1K:~# groups msis
msis : msis users security developers
```